

Efficient and Flexible Spatio-Temporal Transformer for Sequential Visual Place Recognition

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Abstract—Modern sequential Visual Place Recognition (seq-VPR) utilizes transformers to better understand spatio-temporal features in sequential frames. However, existing transformer-based seq-VPR research prioritizes performance over efficiency and flexibility. In application, a seq-VPR model should be (i) efficient in inference speed and memory usage to meet the real-time constraints of VPR systems. It should also be (ii) flexible to the number of frames per sequence (seq-length), since real-world streams seldom provide a fixed seq-length. At present, we believe no existing transformer-based seq-VPR model is both efficient and flexible. To bridge this gap, we introduce Adapt-STformer, built around two key innovations. First, we use a Deformable Transformer Encoder (DTE) that performs sparse attention to learn spatial features. Rather than attending to every other token, each query only attends to a handful of spatial positions, greatly improving (i) computational efficiency compared to spatial attention used in existing seq-VPR methods. Second, at each time step, we feed the previous DTE output and the current frame’s features back into the DTE. Because each step only depends on the last DTE output and one new frame, the model is (ii) flexible to seq-length. We call this combined innovation the Recurrent-DTE module. On Nordland, Oxford, and NuScenes, Adapt-STformer cuts sequence extraction time by 40%, reduces memory by 35%, and improves recall by up to 15% over our strongest baseline, retaining transformers’ performance advantages while delivering both efficiency and flexibility.

Index Terms—Recognition, Localization, SLAM, Visual Place Recognition

I. INTRODUCTION

Sequential Visual Place Recognition (seq-VPR) generates a compact descriptor by aggregating features across sequential frames. By utilizing features across a sequence, seq-VPR methods mitigate the impact of distortions that may only affect individual frames. For example, when the first frame suffers from lighting distortion but subsequent frames remain clear [1], [2]. This temporal fusion also reinforces discriminative features that persist across frames, improving robustness under challenging conditions compared to single-frame VPR methods [3].

Early seq-VPR methods neglect complex cross-frame temporal interactions. For instance, JIST collapses per-frame embeddings into a single descriptor via learnable mean pooling, without direct frame-to-frame interaction [1], [4], [5]. Inspired by advances in video understanding [6]–[8], recent seq-VPR methods integrate transformer modules to model both inter (temporal) and intra (spatial) frame interactions. These methods have yielded impressive SOTA performance [9], [10], especially under challenging VPR conditions.

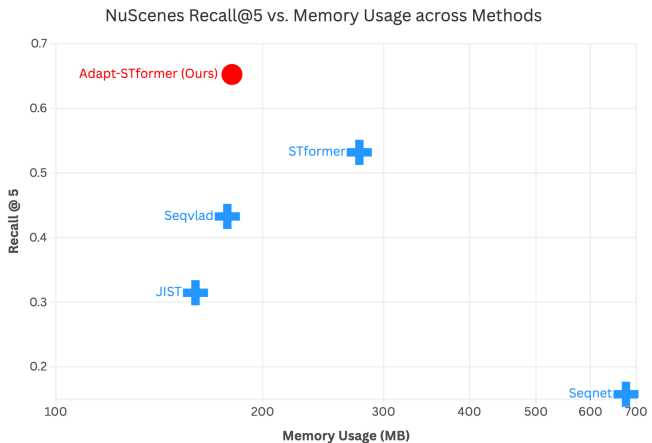


Fig. 1: Adapt-STformer (Ours) utilizes a reasonable amount of memory while achieving higher recall compared to baseline methods on the NuScenes dataset

However, existing transformer-based seq-VPR prioritises research prioritizes performance over efficiency and flexibility. In application, a seq-VPR model must be (i) efficient in inference speed and memory usage to meet the real-time constraints of VPR systems. It must also be (ii) flexible to the number of frames per sequence (seq-length), since real-world streams seldom provide a fixed seq-length. At present, we believe no existing transformer-based seq-VPR model is both efficient and flexible. Approaches like CaseNet [9] utilise a full transformer encoder layer to learn cross-frame temporal interactions. Since each additional frame S generates P tokens, transformer computation scales $\mathcal{O}((S \cdot P)^2)$ making it inefficient. STformer [10] reduces computation by limiting temporal attention to local windows across frames. While this windowing lowers cost, it requires the same seq-length to be used at training and inference. Moreover, STformer keeps separate temporal and spatial encoders, adding significant overhead. Together, these design choices leave STformer neither efficient nor flexible.

To address this gap, we propose Adapt-STformer, a seq-VPR method built around a novel Recurrent Deformable Transformer Encoder (Recurrent-DTE) that combines two key innovations. First, we utilize a DTE [10], [11] to learn spatial features: its deformable attention enables each query to attend

only to a few other tokens, thereby scaling computation roughly linearly rather than quadratically with the token count. Second, at each time step, we feed the previous DTE output and the current frame’s features back into the DTE. Because each step only depends on the last DTE output and one new frame, the model is flexible to sequence length. This recurrent update retains only the previous DTE state and the new frame at each step, enabling direct cross-frame temporal interaction and eliminating the need for a separate transformer layer [12], [13]. Our experiments demonstrate that Adapt-STformer retains transformers’ performance advantages while delivering both efficiency and flexibility.

We summarise our contributions in two points:

- We design, Adapt-STformer, a seq-VPR method well balanced between efficiency, flexibility, and performance (see fig. 1). Achieving a 40 % reduction in sequence extraction time, 35 % memory savings, and up to 15 % recall improvement over our strongest baseline.
- We propose the Recurrent-DTE module, which iteratively feeds the previous deformable-encoder output and the current frame back into the same encoder, unifying sparse spatial attention with recurrent temporal fusion to remain both efficient and flexible.

II. RELATED WORKS

A. Classic seq-VPR

Early methods such as SeqSLAM [14] perform frame-to-frame comparison of query and reference sequences, resulting in high computational costs and relying on a constant-velocity assumption. To address these limitations, SeqNet [1] employs a lightweight 1D temporal convolution to aggregate an entire frame sequence into a single compact descriptor for sequence matching. Improving from Seqnet, SeqVLAD [4] extends NetVLAD [15] by fusing features from all frames into a unified descriptor set and then applying VLAD’s soft-assignment and residual aggregation to yield a fixed-length, seq-length agnostic embedding. Focusing on efficiency, JIST [5] employed SeqGeM, learnable mean pooling, to compress sequences efficiently. These non-transformer-based seq-VPR methods deliver the efficiency and flexibility we value. However, these methods lack the capacity to understand more complex spatio-temporal relationships beyond simple aggregations, thus failing under difficult VPR conditions [10].

B. Transformer-based seq-VPR

STformer [10] is the first fully transformer-based seq-VPR method. It uses a spatial transformer encoder for frame-level features and a sliding window temporal transformer with relative positional encoding to capture the intrinsic dynamics of frame sequences. STformer outperforms earlier non-attention methods and showcases transformers’ potential in seq-VPR. However, it is tied to a fixed sequence length, relies on dense attention, where each query attends to every other token, and maintains separate spatial-temporal transformer encoding modules. CaseNet [9] removes the fixed seq-length constraint

but learns temporal relationships with a full transformer encoder. Transformers’ computations scale quadratically with token count, thus making CaseNet unsustainable with longer sequences. In short, [9], [10] trade flexibility and/or efficiency for performance. Our method aims to restore efficiency and flexibility while maintaining the high performance brought by transformer modules.

C. Deformable Transformer

Deformable Transformer proposed in Deformable DETR [16] replaces dense attention with adaptive sparse sampling. Each query predicts a small set of offsets $\Delta \mathbf{p}$ and weights A via a lightweight projection, easing quadratic cost and slow convergence on high-resolution features. We use this module to learn spatial features.

BEVFormer [2] fuses past BEV features recurrently and uses a deformable transformer encoder for spatial lookup across multi-camera views. We draw on this design, adapting its recurrent fusion and deformable encoding to our seq-VPR method. Xu et al. [17] showed the deformable transformer encoder’s value for multi-camera VPR, but ignored temporal cues. Both methods assume synchronized multi-camera inputs and known intrinsics. We instead target a practical monocular, 2D front-view setting with unknown camera parameters.

D. Challenging VPR

BEV2PR (BEV-Enhanced image VPR) [18] generates explicit BEV structural cues from a single monocular camera by “lifting” RGB features into a pseudo-point cloud and pooling into a BEV map; these BEV features are fused with visual embeddings through dual CNN streams and late-fusion aggregation to improve place recognition under extreme appearance changes (e.g., day vs. night), achieving substantial gains on hard subsets. We draw inspiration from this work, which suggests that focusing on structural cues is beneficial in poor lighting conditions. Motivated by BEV2PR’s use of NuScenes to test extreme appearance changes between day and night, we adopt the NuScenes dataset [19] for our seq-VPR experiments.

METHODOLOGY

A. Overall Architecture

Adapt-STformer takes as input a frame sequence $S = \{s_i\}_{i=1}^L$, with each $s_i \in \mathbb{R}^{H \times W \times 3}$, and outputs a NetVLAD descriptor $V \in \mathbb{R}^{C \times D}$, where C is the number of VLAD clusters and D is the embedding dimension.

There are three intermediate steps between input and output. First, a CCT384 encoder maps the input frame sequence S to a feature tensor $F \in \mathbb{R}^{L \times n \times D}$, where n is the number of tokens per frame and D is the feature dimension. Next, our recurrent deformable transformer encoder refines F in place—preserving its $L \times n \times D$ shape. Finally, our aggregation layer pools it into a compact sequence descriptor $V \in \mathbb{R}^{C \times D}$.

Recurrent-DTE Module: Inspired by [16], we integrate their Deformable Transformer Encoder (DTE) to efficiently learn spatial features. Instead of attending to every token,

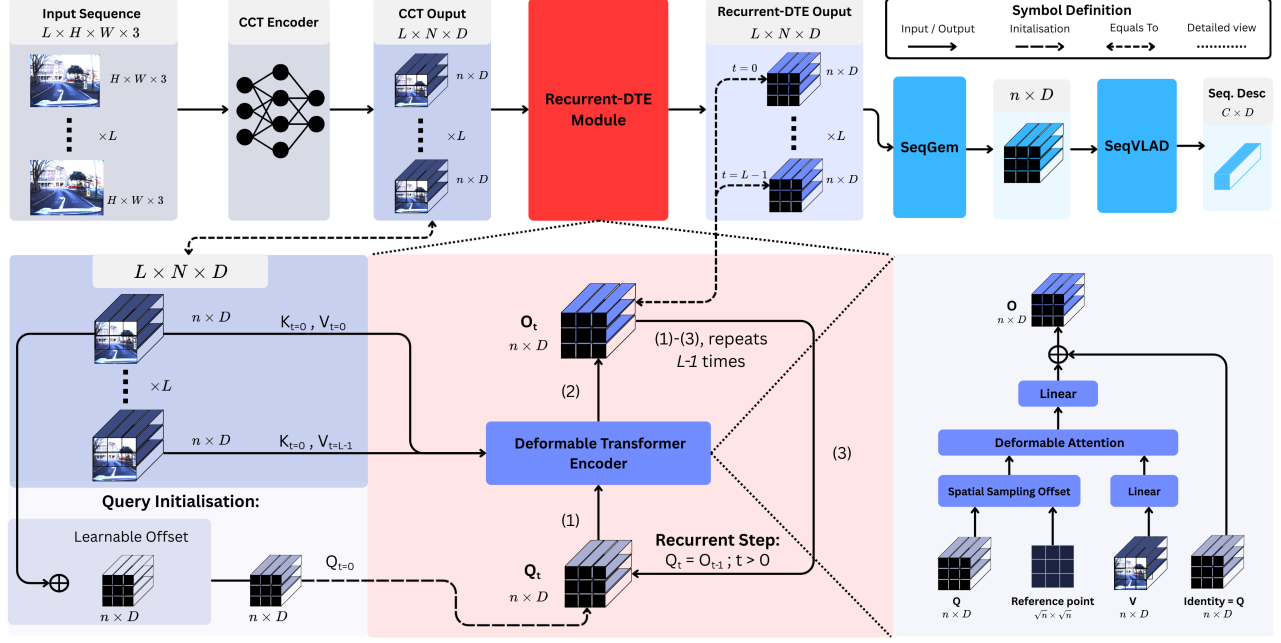


Fig. 2: **Proposed architecture of Adapt-STformer:** An input sequence of frames with dimensions $L \times H \times W \times 3$ is first passed through an image encoder to extract spatial features. At each timestep t , these features act as the keys (K_t) and values (V_t) for our Recurrent Deformable Transformer Encoder (DTE). For $t > 0$, the query (Q_t) is given by the output of the DTE from the previous timestep (O_{t-1}). The DTE then fuses Q_t with (K_t, V_t) to produce the current output (O_t). Outputs across all timesteps are concatenated and further processed by SeqGeM and SeqVLAD, yielding a compact sequential descriptor.

DTE restricts each query to attend to K keypoints around its reference point, which reduces computation from quadratic to near linear complexity. Following [2], [16], we overlay each feature map with a uniform 2D reference point grid and learn per query offsets to dynamically select sampling locations.

To capture temporal dependencies, we introduce a recurrent update: given the frame sequence indices $t = 0, \dots, L-1$, at each timestep t the query Q_t is the DTE output from step $t-1$ and the value V_t is the CCT384 embedding of frame t , i.e.,

$$O_t = \text{DTE}(Q_t = O_{t-1}, V_t),$$

with Q_0 initialized as the first frame’s token embeddings plus a learnable offset. The previous step’s O_{t-1} output acts as a compact state that already encodes all earlier frames; at time t , the DTE fuses this state with the new frame V_t . Because each update depends only on the prior state and the current frame, the same recurrence applies to any number of frames, making the method inherently seq-length agnostic.

Aggregation Module: We save the output from each timestep $\{O_{t=0}, \dots, O_{t=L-1}\}$, to form a tensor of shape $F \in \mathbb{R}^{L \times n \times D}$. We permute F to $F' \in \mathbb{R}^{n \times L \times D}$ (treating tokens as the batch dimension) and apply SeqGeM [5], a

learnable mean-pooling over the temporal axis, collapsing the L dimension to 1:

$$\tilde{F} = \text{SeqGeM}(F') \in \mathbb{R}^{1 \times n \times D}$$

We then feed $\tilde{F}$ into SeqVLAD [4], which aggregates the n D -dimensional embeddings into a descriptor $V \in \mathbb{R}^{C \times D}$, where C is the number of VLAD clusters. Both SeqGeM and SeqVLAD remain agnostic to the input sequence length L .

B. Loss Function

We utilise the same triplet loss detailed in seqnet [1] and in most baseline methods.

EXPERIMENTS

A. Dataset

In our experiments, we use three datasets: NordLand, Oxford RobotCar, and NuScenes. The number of images in train/test is listed in Table I, while the specific filenames of the images used are detailed in the struct files of the codebase.

NordLand: The Nordland dataset [20] comprises a collection of images captured during rail journeys across four seasons, covering various weather and lighting conditions. We

TABLE I: Dataset details specifying the number of images in the database and query sets.

Dataset	Split	Database / Queries
NordLand	Train set	15000 / 15000
	Test set	3000 / 3000
Oxford-Easy	Train set	3619 / 3962
	Test set	3632 / 3921
Oxford-Hard	Train set	2322 / 2585
	Test set	2970 / 2920
NuScenes	Train set	– / –
	Test set	4500 / 4000

use the Summer-Winter pair for training, and the Spring-Fall pair for testing.

Oxford RobotCar: The Oxford RobotCar [21] [22] dataset is a large-scale dataset for autonomous driving research. It encompasses road scenes captured during different periods. We design two experimental sub-datasets: Oxford-easy and Oxford-hard. The Oxford Easy dataset corresponds to the Oxford2 split used in [10]. In Oxford-easy, all 6 sections of the path were used, and 2 meters of frame separation was used. For Oxford-hard, we use a database (2014-11-18-13-20-12, day) and query (2014-12-16 18-44-24, night) for train and database (2014-12-16-09-14 09, day) and query (2014-12-17-18-18-43, night) for test. In Oxford-hard, sections 1,2 of the path were used in training, and sections 3,4 were used in the test set, and a 1-meter frame separation is used. We call it Oxford-Hard because the training and test sequences come from different route segments, and the test set presents more challenging conditions than those encountered during training.

NuScenes: The NuScenes dataset [19] comprises 1,000 scenes, each 20s long, captured at 2 Hz from a full autonomous vehicle sensor suite. We select the Singapore scenes to capture urban day–night traversals under extreme lighting transitions. We split scenes into day and night using scene tags and treat pairs with under a 30° angular difference as positives, following [18]. We use Nuscenes as a test-only dataset, evaluating it on an Oxford-Easy trained model.

B. Model Implementation Details

Model hyperparameters: We use 8 heads, 8 points for the deformable spatial attention, and 2 levels, although we only feed in one feature level.

Training: We train following SeqVLAD [4], initializing from CCT [23] and optimizing with ADAM [24] at a learning rate of 1×10^{-5} . Input images are resized to 384×384 , we use a batch size of 4 (each batch contains one query, one positive, and five negatives), set sequence length $L = 5$, and apply a triplet margin of 0.1. Positives and negatives are mined via GNSS labels: the best positive is the nearest non-trivial neighbor in descriptor space, and negatives are drawn from GNSS-filtered candidates—cached for efficiency—and the hardest are selected. Training stops early if Recall@5 does not improve for five straight epochs.

Evaluation: In the evaluation phase, we use Recall@K as the performance metric. Recall@K is defined as the ratio of the number of correct queries retrieved to the total number of queries. A correct retrieval is defined as at least one of the top K retrievals being within the given radius from the ground truth position of the query. We use radii of 10 meters for Oxford, Nuscene datasets, and 1 frame for the Nordland dataset.

RESULTS

A. Recall comparison

The chosen baseline seq-VPR methods include : SeqNet [1], SeqVLAD [4], STformer [10], and JIST [5]. Their comparative results are summarized in Table II.

Results on easier VPR datasets: SeqVLAD, STformer, and our method achieve near-perfect recall on Nordland. This can be attributed to Nordland’s relatively minor lighting variations, which diminish the visibility of improvements from our lighting-robust design. On the Oxford-Easy dataset, we achieve a +3% recall@1 gain over STformer. While Oxford-Easy presents more lighting challenges than Nordland, its train and test sets share overlapping locations, simplifying the task.

Results on harder VPR datasets: Substantial gains are observed on harder benchmarks: Oxford-Hard (poor lighting, rain, and unseen sections) and NuScenes (poor lighting and cross-domain train-test splits). On Oxford-Hard, we achieve approximately +20% gains in recall@1,5,10 over SeqVLAD. For NuScenes, we observe +12% and +15% improvements in recall@5 and @10, respectively, over STformer, while the recall@1 results are similar.

Other Details: To enable fair comparison with baselines using lower-dimensional descriptors, we apply PCA [25] to model outputs. STformer’s training code is unavailable, and the available pre-trained models were trained on a different data split than our Oxford-Hard set. Therefore, we cannot report its performance on Oxford-Hard. For JIST, we instead evaluate using a publicly available checkpoint, since its original training procedure differs from ours.

B. Resource usage

Computational efficiency is essential for real-world deployment and is a central focus of our work. Table II summarizes the overall resource usage of all methods, while Table III provides a per-module inference time breakdown comparing Adapt-STformer (Ours), STformer (Our main transformer-based seq-VPR baseline), and SeqVLAD. All experiments were conducted in PyTorch [26] on an NVIDIA RTX 8000 GPU.

Transformer breakdown: Deformable attention makes our spatial module 75.8% faster per frame than the dense attention in STformer. Moreover, because Adapt STformer omits a separate temporal transformer encoder, the entire spatio-temporal stage is 96.2% faster than STformer.

Aggregation breakdown: During aggregation, all three methods eventually use SeqVLAD; however, Adapt STformer first applies SeqGeM that collapses the $L \times n \times D$ tensor to

TABLE II: Recall performance of models and resource usage. The best overall recall in each dimension group on each dataset is indicated using **green**, while the second-best results are underlined.

Dimension	Method	Backbone	Nordland			Oxford-Easy			Oxford-Hard			NuScenes			Resource Usage		
			R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	Mem (MB)	Parameters	Time (s)
24576	Seqvlad	CCT384	0.9603	0.9947	<u>0.9947</u>	0.8275	0.9249	0.9541	<u>0.4709</u>	<u>0.5601</u>	<u>0.6151</u>	0.3236	0.4327	0.5142	177.66	13.3M	0.0222
	STformer	CCT384	<u>0.9697</u>	0.9947	0.9950	<u>0.8488</u>	<u>0.9323</u>	<u>0.9633</u>	–	–	–	<u>0.4461</u>	<u>0.5319</u>	<u>0.5945</u>	276.55	13.5M	0.0338
	Our method	CCT384	0.9763	<u>0.9943</u>	<u>0.9947</u>	0.8854	0.9528	0.9700	0.6940	0.7706	0.7969	0.4500	0.6524	0.7421	180.39	13.9 M	0.0206
4096	Seqnet	NetVLAD	0.7943	0.9013	0.9317	0.5363	0.7467	0.8241	0.3064	0.4505	0.5228	0.0827	0.1575	0.2031	676.09	83.9M	0.0042
	Seqvlad (PCA)	CCT384	0.9567	0.9947	<u>0.9947</u>	0.8280	0.9251	0.9557	<u>0.4603</u>	<u>0.5550</u>	<u>0.6151</u>	0.3236	0.4327	0.5146			
	STformer (PCA)	CCT384	0.9700	0.9947	0.9950	<u>0.8534</u>	<u>0.9372</u>	<u>0.9667</u>	–	–	–	<u>0.4461</u>	<u>0.5319</u>	<u>0.5949</u>			
	Our method (PCA)	CCT384	<u>0.9610</u>	<u>0.9923</u>	<u>0.9937</u>	0.8882	0.9544	0.9710	0.6874	0.7659	0.7918	0.4500	0.6563	0.7413			
512	JIST	ResNet	0.8210	0.9200	0.9450	0.5868	0.7416	0.8134	<u>0.4423</u>	<u>0.5405</u>	<u>0.6139</u>	0.2047	0.3146	0.3713	159.65	11.4M	0.0077
	Seqvlad (PCA)	CCT384	<u>0.9567</u>	0.9947	<u>0.9947</u>	0.8236	0.9244	0.9564	0.4383	0.5330	0.6049	0.3146	0.4236	0.5024			
	STformer (PCA)	CCT384	0.9637	<u>0.9933</u>	0.9950	<u>0.8513</u>	<u>0.9369</u>	<u>0.9667</u>	–	–	–	<u>0.4445</u>	<u>0.5370</u>	<u>0.5969</u>			
	Our method (PCA)	CCT384	0.9490	0.9900	0.9943	0.8870	0.9554	0.9731	0.6516	0.7459	0.7765	0.4390	0.6386	0.7382			

TABLE III: Per-module inference times for each method. Fastest time per module and speedup percentages are in **blue**.

Method	Spatial Attn	Spatio-Temporal	Aggregation	Overall
SeqVLAD	–	–	0.00394s	0.02142s
STformer	0.00182s	0.02535s	0.00518s	0.03440s
Our Method	0.00044s	0.00096s	0.00154s	0.02001s
% Faster vs STformer	-75.8%	-96.2%	-70.3%	-41.8%
% Faster vs SeqVLAD	–	–	-60.9%	-6.6%

TABLE IV: Ablation study evaluating sequence length L and temporal method’s impact on performance and resources.

L	Temporal Method	Oxford-Hard R@5	NuScenes R@5	Mem (MB)	Time (s)
1	Transformer	0.7368	0.4827	106.02	0.0129
	seqGeM only	0.7306	<u>0.5154</u>	88.71	0.0113
	Ours	<u>0.7341</u>	0.6224	88.71	0.0117
3	Transformer	<u>0.7384</u>	0.4520	152.03	0.0202
	seqGeM only	0.7255	<u>0.4622</u>	134.39	0.0134
	Ours	0.7549	0.6051	134.39	0.0137
5	Transformer	<u>0.7341</u>	<u>0.4480</u>	198.16	0.0319
	seqGeM only	0.7247	0.4201	180.39	0.0197
	Ours	0.7706	0.6559	180.39	0.0206

$1 \times n \times D$. Since SeqGeM adds virtually no overhead, this extra step shortens the subsequent SeqVLAD operation, lowering SeqVLAD ggregation time by 60.9% relative to SeqVLAD and by 70.3% relative to STformer.

Overall comparison: Relative to STformer, Adapt-STformer extracts descriptors 40% faster while using roughly 35% less memory per sequence, yielding an end-to-end extraction time that is still lower than SeqVLAD, a non-transformer seq-VPR method.

TABLE V: Per-module inference times for each method. Fastest time per module and speedup percentages are in **blue**.

Attention Type	Oxford Hard	NuScenes	Mem (MB)	Time(s)
Dense	0.6858	0.3138	182.75	0.0208
Deformable (Ours)	0.7706	0.6559	180.39	0.0197
% Improvement	+8.48%	+34.21%	-1.30%	-5.39%

C. Ablation Study

We conduct an ablation study on the Oxford and Nuscenes datasets to evaluate the impact of sequence length (L), compare temporal fusion techniques (Transformer vs SeqGeM only vs Recurrent + SeqGeM (ours)), and isolate the impact of the Deformable Transformer Encoder module. All models are trained on sequence length $L = 5$ but evaluated at $L = 3$ and $L = 1$, demonstrating their ability to handle shorter seq-lengths when inference resources are limited dynamically [27].

Sequence Length: As shown in Table IV, our recurrent+SeqGeM temporal method’s R@5 on Oxford-Hard rises from 0.7341 ($L = 1$) to 0.7549 ($L = 3$) and 0.7706 ($L = 5$). On NuScenes, it likewise increases from 0.6224 to 0.6051 and 0.6559. These improvement shows that our temporal fusion module is effective at using extra information available in longer seq-lengths.

Temporal Fusion Techniques: Our Recurrent + SeqGeM fusion beats only SeqGeM at every sequence length, raising R5 on both Oxford-Hard and NuScenes by several to double-digit percentages while adding no extra memory usage and negligible increase in extraction time. Against the Transformer encoder, our approach matches or outperforms its recall while using far fewer resources. For example, at $L = 3$ we record 0.6051 vs. 0.452 on NuScenes, yet we run 32 % faster (0.0137s vs. 0.0202s) and consume 12 % less memory (134.39MB vs. 152.03MB). These results highlight that our recurrent fusion promotes direct cross-frame interactions, enabling it to outperform naive aggregation (SeqGeM only) while requiring far less compute than transformer encoders.

DTE module: To evaluate the effectiveness of deformable attention for spatial feature learning, we replace our DTE with a dense attention-based spatial transformer encoder, similar to that in STformer [10]. As shown in Table V, our deformable design improves recall on Oxford Hard +8.48% and NuScenes +34.21% while also reducing memory usage –1.30% and inference time –5.39%. Although the dense attention encoder may not align perfectly with our pipeline, the results highlight that DTE achieves lower computational cost than dense alternatives

while maintaining strong performance.

D. Qualitative Results

Based on the results above, DTE is effective in learning spatial features. To illustrate this, we visualize attention maps from our method and from the baselines. In Figure 3 (a)(b) and (c)(d), two NuScenes examples show our model retrieving correctly while STformer fails. In both day and night scenes, STformer produces smooth, concentrated activations over only a few regions. By contrast, our model displays many scattered high-activation patches across the entire frame. This difference arises from the type of attention used. Global dense attention computes similarity between every pixel pair, which often results in blob-shaped activations around the most informative regions. Deformable attention instead learns a small set of sparse offsets that jump directly to features relative to the query token, producing scattered activations. Under poor lighting, many features are weak. In this case, capturing any available geometric cue becomes crucial for recall. This is similar to how a human might navigate in the dark by noticing any visible structure rather than relying on regions that were informative in daylight. We observe the same pattern in Figures 4 (a)(b), and (c)(d) when comparing our method to SeqVLAD activation maps on the Oxford Hard dataset.

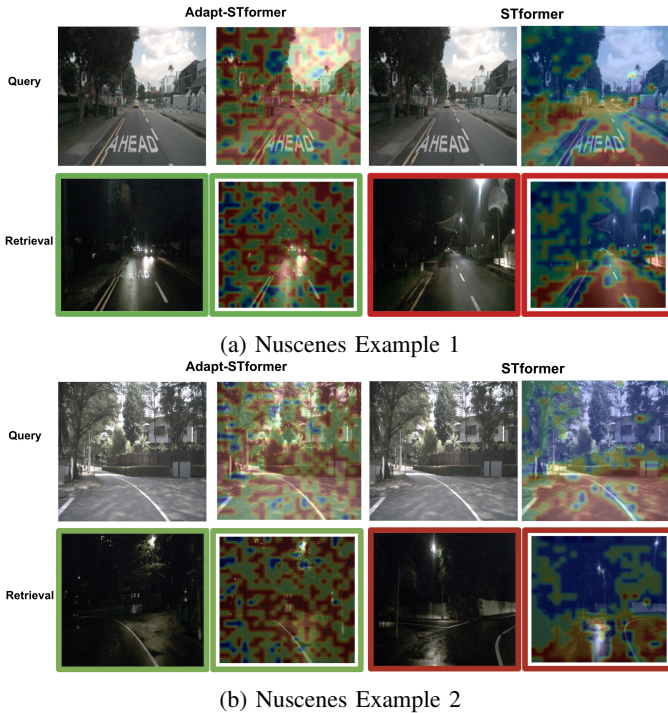


Fig. 3: Qualitative comparisons across NuScenes

E. Limitations

Our model is not yet able to fully take advantage of the most powerful VPR backbones, such as the DINOv2 variants [28] used in [9], [29], and therefore, we are not confident in claiming global SOTA recall results. We currently employ CCT384 [23], a hybrid convolutional and transformer

backbone used in [4], [10], which offers greater computational efficiency than pure ViT [30] backbones such as DINOv2.

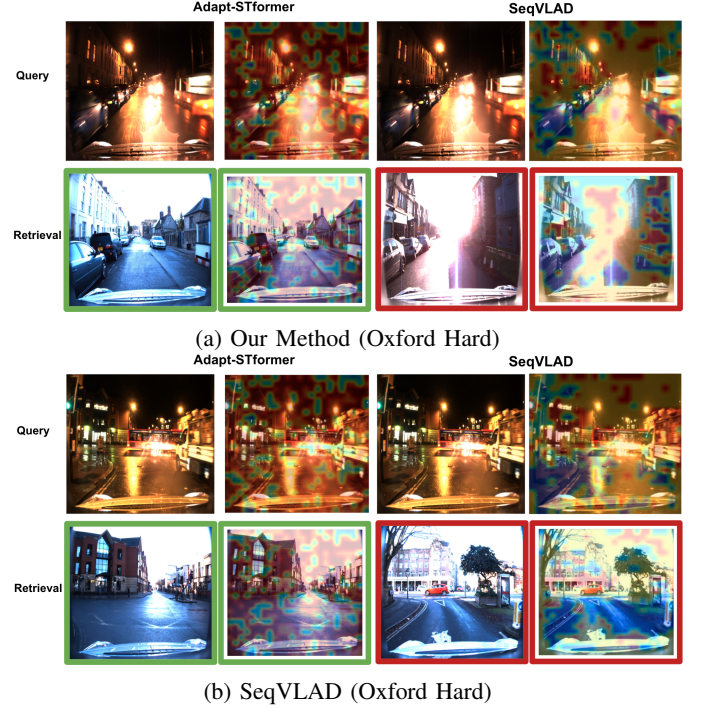


Fig. 4: Qualitative comparisons across oxford-hard

III. CONCLUSION

We have shown that transformer-based seq-VPR models can be re-designed to deliver both efficiency and flexibility. By combining sparse spatial attention with a recurrent temporal update, Adapt-STformer narrows the gap between high performance and practical use. As seq-VPR advances, we encourage designs that push performance while staying efficient, adaptable, and practical for real-world localization.

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